SOFTWARE ENGINEERING LAB 1

Functional and Non-Functional Requirements

Project Title	Student Club Event Ticketing & Budget Portal
Primary Domain	Campus & Academic Operations
Primary Actors	Club Lead, Campus Admin
Name	I.V. Thanvitha
SRN	PES2UG24CS197

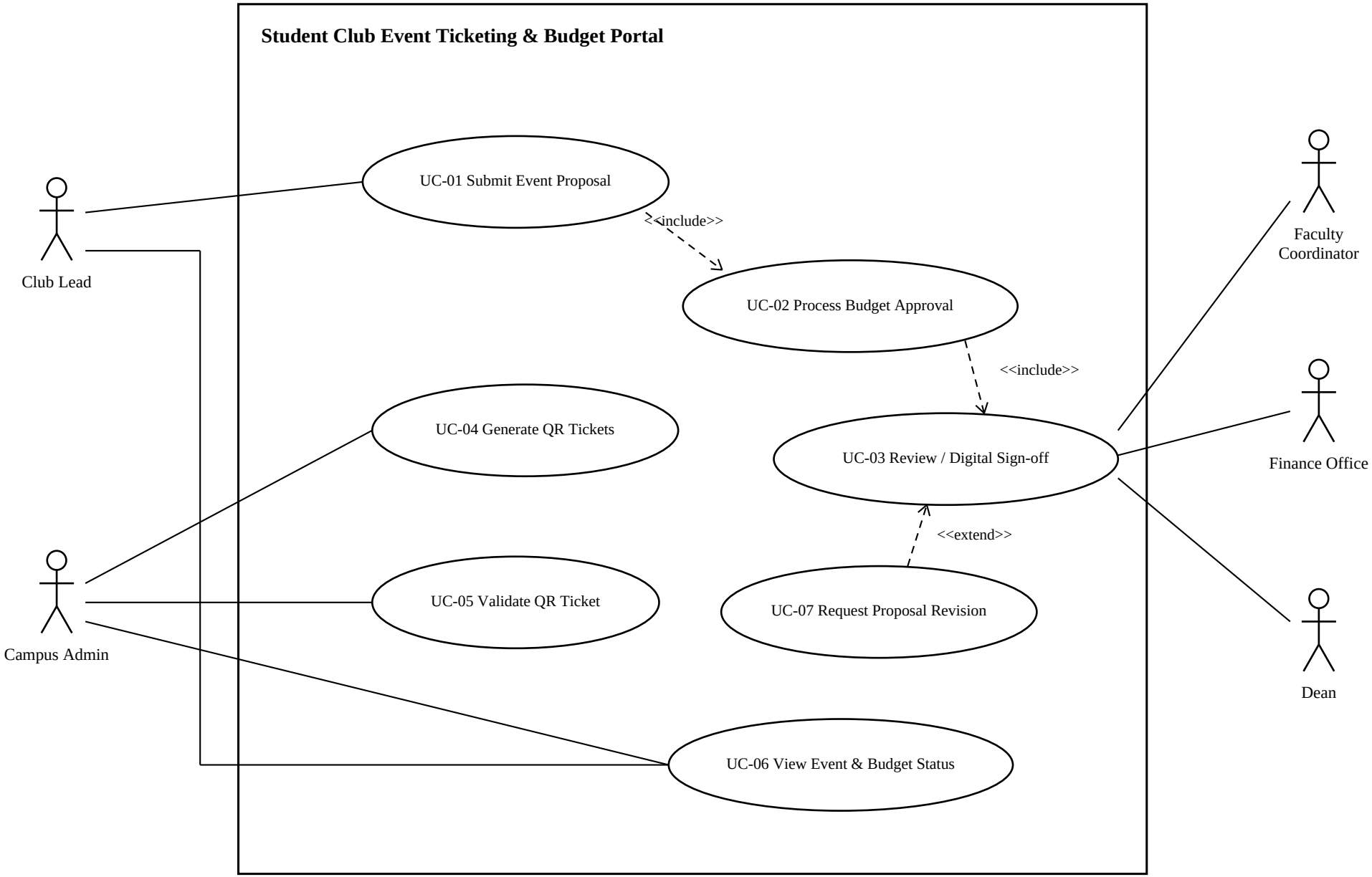
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow in the sequence Faculty Coordinator -> Finance Office -> Dean.	High	PASS: A submitted proposal moves to the next stage only after the current stage records a digital sign-off. FAIL: The proposal reaches the Dean before Finance Office sign-off, or any stage is skipped.	Enforces the university approval hierarchy and prevents unauthorized budget progression.
FR-002	Functional	The system shall allow a Club Lead to create and submit an event proposal containing event details, requested budget, and planned ticket quantity.	High	PASS: With all mandatory fields completed, submission creates a unique proposal ID and status "Submitted". FAIL: Submission succeeds when any mandatory field is missing or invalid.	Captures the minimum information needed to evaluate an event and its funding request.
FR-003	Functional	The system shall allow the Club Lead and Campus Admin to view the current approval stage, decision history, approved budget amount, and ticketing status for each event proposal.	Medium	PASS: After each approval decision, the portal displays the correct current stage and decision history within 2 seconds of refresh. FAIL: The displayed stage/history differs from the stored workflow state.	Provides traceability and reduces manual follow-up between clubs and campus administration.
FR-004	Functional	The system shall generate a unique QR-code ticket for each attendee allocation only after the event has received final approval.	High	PASS: An approved event can generate the requested ticket quantity and 1,000 generated test tickets contain no duplicate ticket IDs. FAIL: Ticket generation is allowed before final approval or duplicate ticket IDs are produced.	Ensures tickets are tied to approved events and can be uniquely verified at entry.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall validate a presented QR-code ticket at venue check-in and shall reject invalid, wrong-event, or previously used tickets while recording successful check-ins.	High	PASS: A valid unused ticket is accepted once and marked "Checked-in"; a second scan, altered code, or ticket for another event is rejected. FAIL: Any duplicate or invalid ticket is accepted.	Prevents ticket reuse and unauthorized entry while maintaining an attendance record.
NFR-001	Nonfunctional - Performance & Security	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	PASS: In a simulated peak-load benchmark, every test validation response completes in <= 100 ms and altered/invalid ticket data is rejected. FAIL: Any validation exceeds 100 ms or invalid data is accepted.	Supports fast queue movement at entry points without weakening ticket authenticity checks.
NFR-002	Nonfunctional - Security & Auditability	The system shall enforce role-based access control for Club Lead, Campus Admin, Faculty Coordinator, Finance Office, and Dean actions, and shall maintain an audit record for approval and ticket-administration actions.	High	PASS: 20/20 unauthorized-role access tests are denied, and 100% of approval/ticket-administration test actions are logged with actor role, action, timestamp, and affected record. FAIL: Any unauthorized action succeeds or any required audit entry is missing.	Protects privileged operations and provides evidence for administrative review.

Requirement quality check: Each requirement uses a "shall" statement, has a priority, and has measurable pass/fail acceptance criteria.

UML Use-Case Diagram

Student Club Event Ticketing & Budget Portal



Primary actors: Club Lead, Campus Admin. Approval actors are included because FR-001 requires Faculty Coordinator -> Finance Office -> Dean sign-off.

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Use-Case Flow Specification

Project	Student Club Event Ticketing & Budget Portal
Use Case	UC-02 - Process Multi-Stage Budget Approval
Primary Actor(s)	Faculty Coordinator, Finance Office, Dean
Supporting Actor(s)	Club Lead; Campus Admin (status visibility)
Preconditions	Club Lead is authenticated; an event proposal with a requested budget has been submitted; the proposal is in Pending Approval state.
Postconditions	The proposal is Approved, Rejected, or Returned for Revision; every decision is recorded. If fully approved, the approved budget is stored and QR-ticket generation becomes eligible.

Main Success Scenario

1. The Club Lead submits the completed event proposal and requested budget.
2. The system assigns a proposal ID, sets the status to "Pending - Faculty Coordinator", and routes the proposal to the Faculty Coordinator.
3. The Faculty Coordinator reviews the proposal and records a digital approval.
4. The system records the sign-off with timestamp and routes the proposal to the Finance Office.
5. The Finance Office reviews the requested amount and records a digital approval.
6. The system records the finance sign-off and routes the proposal to the Dean.
7. The Dean reviews the fully signed proposal and gives final approval.
8. The system records the Dean sign-off, changes the proposal status to "Approved", and stores the approved budget amount.
9. The system enables QR-ticket generation for the approved event and updates the event/budget status view.
10. The system notifies the Club Lead and Campus Admin that the approval workflow is complete.

Alternate Flow - 5a: Finance Office Requests Revision

5a1. The Finance Office selects "Request Revision" and enters a mandatory reason/comment.

5a2. The system records the decision, changes the status to "Revision Required", and does not route the proposal to the Dean.

5a3. The system notifies the Club Lead and displays the revision comments.

5a4. The Club Lead edits the proposal and resubmits it; the system restarts the sequential approval workflow from the Faculty Coordinator.